Single-Agent Foundations

Kernelised bandits and misspecified linear bandits

CS6007 project notes

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What this folder is about. Two single-agent papers that motivate and benchmark single-index bandits: one handles a fully *nonparametric* reward (kernels), the other shows what goes wrong when a *linear* reward is assumed but false. Both are by authors close to our project (A. Gopalan; A. Ghosh is our instructor).

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1 Chowdhury & Gopalan (2017): kernelised multi-armed bandits

S. R. Chowdhury, A. Gopalan. On Kernelized Multi-armed Bandits. ICML 2017 (arXiv:1704.00445).
File: `arxiv-1704.00445.pdf`

Setting.

- Maximise an unknown function f in a reproducing kernel Hilbert space (RKHS) from noisy evaluations; no parametric form.

What they did.

- **IGP-UCB**: an improved GP-UCB whose confidence width is smaller by roughly $\ln^{3/2} t$.
- **GP-TS**: a GP-based Thompson sampling algorithm.

Main results.

- Regret bounds in terms of the information gain γ_T of the kernel; IGP-UCB improves GP-UCB in theory and practice.
- The first frequentist regret bound for Thompson sampling on fixed, unknown RKHS functions.

Why it matters for our project.

- IGP-UCB is one of our single-agent baselines (E2, E4): it degrades with dimension d because γ_T grows with d, which single-index structure avoids.
- Coop-KernelUCB (folder 04) is built on its confidence bounds.

Open problems.

- **Stated by the authors:** Learn the kernel adaptively when it is unknown.
- **Stated by the authors:** Time-varying functions from the RKHS.
- **Stated by the authors:** Reinforcement learning with GP techniques.
- **Stated by the authors:** Computational cost of optimising functions drawn from GPs.

2 Ghosh, Chowdhury & Gopalan (2017): misspecified linear bandits

A. Ghosh, S. R. Chowdhury, A. Gopalan. Misspecified Linear Bandits. AAAI 2017 (arXiv:1704.06880).
File: `arxiv-1704.06880.pdf`

Setting.

- Linear bandits where the true reward deviates from linearity (sparse deviations).

What they did.

- Shows OFUL, a state-of-the-art linear bandit algorithm, is not robust to such deviations.
- Proposes **RLB**: test for linearity on the fly and fall back to a non-parametric UCB when the test fails.

Main results.

- A formal regret upper bound for RLB; experiments (synthetic and real) show real feature-to-reward maps are often far from linear.

Why it matters for our project.

- The motivation for unknown links: assuming the wrong model can give linear regret. Dey et al. cite it for exactly this.

Open problems.

- **Stated by the authors:** Other non-linearity structures than sparse deviations.
- **Stated by the authors:** Information-theoretic regret lower bounds for general bandits with given feature sets.
- **Stated by the authors:** Thompson sampling on linear bandits with deviations.

3 Open problems across this folder

- Both papers pay for flexibility: kernels pay in dimension (through γ_T), robust tests pay by falling back to unstructured UCB. Single-index models sit between them.
- Neither studies multiple agents; multi-agent versions of both exist (Coop-KernelUCB in folder 04).